Final Project & Presentation: Choose Your Own Task and Create and Deliver a Presentation

Prerequisite Work: At this point, you have learned quite a bit of Java programming. You have taken the APCSA test, and hopefully you can feel a little bit of relief now. However, your year is not finished just yet.

Project Description: You will be given some options as to what you can do for your final project. I have compiled a few options below, but this list is not exhaustive.

- **Try a Kaggle course on any of the following topics:**
  - Machine Learning (Intro/ Intermediate)
  - Data Visualization
  - Pandas
  - SQL (Intro / Advanced)
  - Intro to AI Agents
  - Computer Vision
  - Data Cleaning
  - Natural Language Processing
  - or others …
- **Try one of the following CodeHS courses:**
  - Mobile Apps in React Native
  - Data Science with Python
  - Introduction to Artificial Intelligence
  - Introduction to C++
  - Data Structures in C++
  - Introduction to Physical Computing with Arduino
  - Introduction to SQL
- **Create an informative magazine or children's book on any unit we learned about this year**
  - Check out the Bubble Sort Zines or Wizard Zines as an example. You can make it by hand or digitally and print it out.
- **Learn p5.js, a free, open source, visual programming language based on JavaScript**
  - Here are some project ideas. And a few more ideas.
  - Here is the CodeAcademy p5.js course
- **Continue learning Java in by:**
  - Learning about Abstract Classes and Interfaces. I can put some materials on CodeHS for this.
- **MIT App Inventor Or Code.org App Lab**
  - Use MIT App Inventor's Getting Started Guide, tutorials, and other tools to create a mobile app of your own. .
- **EarSketch:** Program music using Python
  - Use Ear Sketch's Hour of Code (directions on right side), tutorials, and videos to create music through code.
  - Check it out here
  - Read the getting started guide here
- **FreeCodeCamp:** a tiny non-profit that's helping millions of people learn to code for free.
  - Start one of the 7 curriculums offered
- **Harvard Intro CS Course on EdX:** Plenty of options to choose from. They are free and you don't have to totally complete it to get credit on your final project.
  - CS50's Introduction to Game Development
  - CS50's Mobile App Development with React Native
  - CS50's Web Programming with Python and JavaScript
  - CS50: Introduction to Computer Science
  - CS50's Introduction to Artificial Intelligence with Python
- **Complete parts of this course: The Missing Semester of Your CS Education**
- **Take a free online AI course, such as Elements of AI, CS50s AI with Python, or Practical Deep Learning for Coders**
- **Complete one or more of the following labs:**
  - Meteorite Landing Data Analysis with Java FX: **https://learn.java/learning/mini-labs/UsingRealDataMiniLab**
  - Veterinarian Office - Pattern Matching Lab: **https://learn.java/learning/mini-labs/PatternMatchingMiniLab**
  - Card Game: **https://learn.java/learning/mini-labs/culminating/card-game**
- **Write a program for the Congressional App Challenge**
- … Or **another CS-related project** that you can articulate that is appropriate for the class and is approved by the instructor.

Use of AI on this assignment

Because AI tools are becoming increasingly important in computer science and software development, you may choose to thoughtfully incorporate AI tools into your project workflow.

AI tools are allowed on this assignment when used appropriately and honestly.

The purpose of this project is for you to learn, explore, create, and demonstrate understanding. AI should support your learning — not replace it.

Appropriate AI Use

Examples of acceptable AI use include:

- asking AI to explain concepts
- debugging code
- generating small examples or starter code
- brainstorming ideas
- improving wording or organization
- helping learn unfamiliar syntax/tools

Inappropriate AI Use

Examples of unacceptable AI use include:

- copying large amounts of AI-generated code without understanding it
- submitting AI-generated work as entirely your own
- using AI to complete the entire project with minimal personal contribution
- generating a presentation/script that you cannot explain yourself
- using AI in ways that bypass the learning process

Transparency Requirement

If you use AI tools during your project, you should:

- briefly mention which AI tools you used
- explain how they helped you
- describe what YOU personally changed, learned, or created

This can be included in:

- your daily logs,
- presentation,
- or final reflection.

Final Expectation

At the end of the project, you should still be able to honestly say:

“I understand this project, I can explain how it works, and I made meaningful contributions myself.”

Using AI responsibly is a valuable skill. Simply copying from AI without understanding is not.

Project Parameters:

<u>Planning: Picking Your Project & Setting a Reasonable Goal</u>

You will need to choose your project. You will need to choose the goal that you will work towards. You will be responsible for a final product as well as a presentation of the final product. Please make sure your goal is both reasonable and achievable.

> *An example of a good goal looks like this:*
> "I plan to create a (simple to moderately advanced) mobile app using MIT's App Inventor. In my final presentation, I will present my learning and experiences in App Inventor, show my classmates what it is and how to use it, and unveil my individual application to the class."

Poor goal designing usually results from being too ambitious (picking a project goal that needs way more time than you have) or not ambitious/specific enough (too simple or broad).

<u>Documentation:</u>
You should document the things you are learning and experiencing throughout the project time. Every day, you will complete a daily progress check on Canvas, answering the following questions:

- What did you do/learn?
- What did you accomplish?
- What difficulties did you face? How did you overcome them?
- What will you do next?

<u>Final Product:</u>
Part of your goal should include some sort of final product. That can be anything from multiple tutorials completed in Processing, an original song written in Earsketch, etc.. This final product should be something that you made throughout the time given. It doesn't have to be completely original in all cases, especially if done with the help of a tutorial or AI, but you should definitely spend some time making it "yours."

When you finish your final product(s), you should be able to say "I made this myself with some help from a tutorial." You should not be submitting work that was already made or was written to teach you. You also should not submit work that was created by AI. You should edit or expand upon anything that you find enough to make it yours.

<u>Presentation:</u>
You will make a 4-6 minute presentation in which you describe your project and present your final product.

In your presentation, you should present and address:
- Your goal
- Platform/language used and a solid description on what the platform/language is
- What you learned throughout the given work time, focusing on overall key learning points, your biggest accomplishments, and biggest difficulties faced
- Reflection of your project and assessment of achieving your goal. Are you happy with the outcome? Would you recommend it to others to try? Why or why not?
- A demo of your work products

<u>Timeline</u>**:**
- **Tuesday, May 19:** Enter project idea and goal to document on Canvas.
- **Tuesday, May 19 – Friday, May 29:** Begin daily work on your project. Every class day requires submission to the daily progress form.
- **Monday, June 1, 4 PM (Wednesday, May 27, 4 PM for seniors):** Submit presentation (use Slides, Canva, or a similar tool) and your final product to Canvas.
- **Presentations** will be Wednesday, May 27 for seniors and Tuesday-Thursday June 2-4 for other students

Rubric:

Your grade will be earned through a combination of your use of class time & productivity, the quality of your presentation, and the appropriateness and success to achieving your goal. This will count as a test grade.

Project Work Time (20 points based on daily form completion and appropriate use of class time)
Presentation (5-10 minutes in length) (15 points)
Final Products (Appropriate & Working) (15 points)
Total Points for Project: 50

Note that you will lose 5 points on the project for each progress check/project description that is not turned in within two days of its due date or for each day that you choose to play computer games during class.

Plagiarism: Taking a significant piece of code, entire program or project that was not created by yourself, and submitting it as your project would be an honor code violation.

Criteria	**Excellent/Well Done**	**Average/Decent**	**Needs Work**
Work Time	Daily logs of learned concepts, accomplishments, and difficulties for each day are submitted on time and are highly detailed. **16-20**	Daily logs of learned concepts, accomplishments, and difficulties for each day are submitted and are moderately detailed. **11-15**	Daily logs of learned concepts, accomplishments, and difficulties for each day are submitted and lack much detail. **0-10**
Presentation	The presentation included and thoroughly addressed all criteria listed in the presentation section of this document. The presentation is 4 -6 minutes in length. The presentation is well balanced with text and photos **11-15**	The presentation included and addressed most criteria listed in the presentation section of this document. The presentation is up to one minute inside or outside the time limit. The presentation is well balanced with text and photos **6-10**	The presentation included and addressed some criteria listed in the presentation section of this document. The presentation was too short. **0-5**
Final Product(s)	Final products are displayed in the slideshow with either screenshots or videos. The final product and the concepts needed to complete it are explained in detail. **11-15**	Final products are displayed in the slideshow with either screenshots or videos. The final product and the concepts needed to complete it are generally explained. **6-10**	Final products are displayed in the slideshow with either screenshots or videos. **0-5**